

6.4

1) Find the period.

$$y = -5 \tan(3\pi x)$$

$$y = -5 \tan(3\pi(x))$$

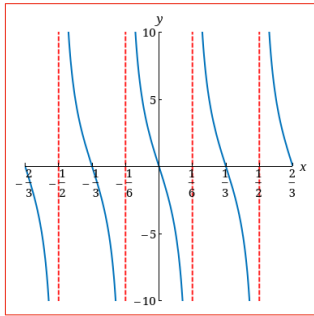
$$P: \frac{\pi}{k} = \frac{\pi}{3\pi} = \boxed{\frac{1}{3}}$$

$$\text{Interval: } (-\frac{\pi}{2k} + b, \frac{\pi}{2k} + b)$$

$$(-\frac{\pi}{6\pi} + 0, \frac{\pi}{6\pi} + 0)$$

$$(-\frac{1}{6}, \frac{1}{6})$$

Graph



2) Find the period

$$y = 2 \csc(x + \frac{\pi}{2})$$

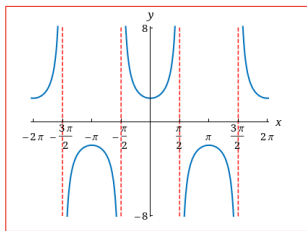
$$P: \frac{2\pi}{1} = \boxed{2\pi}$$

$$\text{Interval: } (0 + b, \frac{2\pi}{k} + b)$$

$$(0 + (-\frac{\pi}{2}), \frac{2\pi}{1} + (-\frac{\pi}{2}))$$

$$(-\frac{\pi}{2}, \frac{3\pi}{2})$$

Graph



3) Find the period

$$y = 3 \cot(2x - \frac{\pi}{2})$$

$$y = 3 \cot(2(x - \frac{\pi}{4}))$$

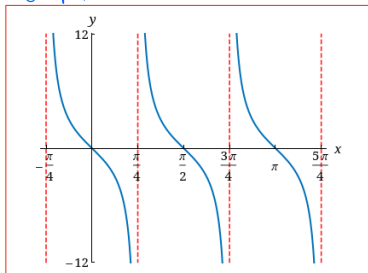
$$P: \frac{\pi}{k} = \boxed{\frac{\pi}{2}}$$

$$\text{Interval: } (0 + b, \frac{\pi}{k} + b)$$

$$(0 + \frac{\pi}{4}, \frac{\pi}{2} + \frac{\pi}{4})$$

$$(\frac{\pi}{4}, \frac{3\pi}{4})$$

Graph



4) Find the period

$$y = \frac{1}{4} \sec(2\pi x - \pi)$$

$$y = \frac{1}{4} \sec(2\pi(x - \frac{1}{2}))$$

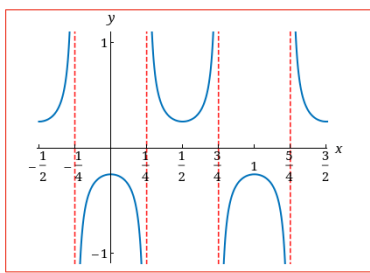
$$P: \frac{2\pi}{k} = \frac{2\pi}{2\pi} = \boxed{1}$$

$$\text{Interval: } (0 + b, \frac{2\pi}{k} + b)$$

$$(0 + \frac{1}{2}, 1 + \frac{1}{2})$$

$$(\frac{1}{2}, \frac{3}{2})$$

Graph



5) Find the period

$$y = 4 \tan\left(\frac{1}{2}\left(x + \frac{\pi}{4}\right)\right)$$

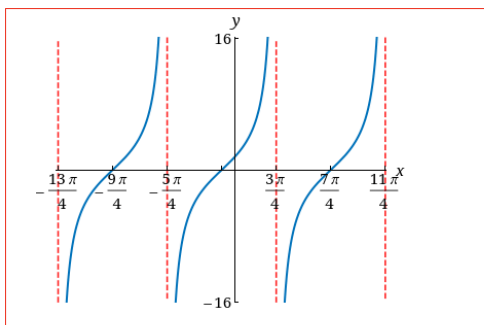
$$p: \frac{\pi}{k} = \frac{\pi}{\frac{1}{2}} = \boxed{2\pi}$$

Interval: $\left(-\frac{\pi}{2k} + b, \frac{\pi}{2k} + b\right)$

$$\left(-\pi + \left(-\frac{\pi}{4}\right), \pi + \left(-\frac{\pi}{4}\right)\right)$$

$$\left(-\frac{5\pi}{4}, \frac{3\pi}{4}\right)$$

Graph



6.5

1) (a) $\cos^{-1}(-1)$

$$\boxed{180^\circ}$$

(b) $\cos^{-1}\left(\frac{1}{2}\right)$

$$\boxed{60^\circ}$$

(c) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

$$\boxed{150^\circ}$$

2) Use a calculator to find

$$\tan^{-1}(-24)$$

$$\approx \boxed{1.52915}$$

3) $\cos(\cos^{-1}\left(\frac{5}{8}\right))$

$$\cos^{-1}: [-1, 1]$$

$$\boxed{\frac{5}{8}}$$

4) $\sin(\sin^{-1}(2))$

$$\sin^{-1}: [-1, 1]$$

$$\boxed{\text{UNDEFINED}}$$

5) $\cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right)$

$$\cos^{-1}: [0, \pi] \rightarrow \text{QI}$$

$$\boxed{\frac{\pi}{3}}$$

6) $\cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)$

$$\cos^{-1}: [0, \pi] \rightarrow \text{QIII}$$

$$\cos^{-1}\left(-\cos\left(\frac{\pi}{4}\right)\right)$$

$$= \cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$$

$$= \boxed{\frac{3\pi}{4}}$$

$$7) \tan^{-1}\left(\tan\left(-\frac{\pi}{6}\right)\right)$$

$$\tan^{-1}: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\boxed{-\frac{\pi}{6}}$$

$$8) \tan\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)$$

$$\sin^{-1}: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\tan\left(\frac{\pi}{3}\right)$$

$$= \boxed{\sqrt{3}}$$

$$9) \sin^{-1}\left(\sin\left(\frac{4\pi}{3}\right)\right)$$

$$\sin^{-1}: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \xrightarrow{+\pi}$$

$$= \sin^{-1}\left(-\sin\left(\frac{\pi}{3}\right)\right)$$

$$= \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$$

$$= \boxed{-\frac{\pi}{3}}$$

$$10) \sec\left(\tan^{-1}(-1)\right)$$

$$\tan^{-1}: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$= \sec\left(-\frac{\pi}{4}\right)$$

$$= \frac{1}{\cos\left(-\frac{\pi}{4}\right)}$$

$$= \frac{1}{\cos\left(\frac{\pi}{4}\right)}$$

$$= \frac{1}{\frac{\sqrt{2}}{2}}$$

$$= \boxed{\sqrt{2}}$$

$$11) \csc\left(\tan^{-1}(-1)\right)$$

$$\tan^{-1}: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\csc\left(-\frac{\pi}{4}\right)$$

$$= \frac{1}{-\sin\left(\frac{\pi}{4}\right)}$$

$$= \frac{1}{-\frac{\sqrt{2}}{2}}$$

$$= \boxed{-\sqrt{2}}$$

$$12) \tan\left(\cos^{-1}\left(-\frac{1}{2}\right)\right)$$

$$\cos^{-1}: [0, \pi]$$

$$\tan\left(\frac{2\pi}{3}\right)$$

$$= \boxed{-\sqrt{3}}$$

$$13) \cos\left(\sin^{-1}(-1)\right)$$

$$\sin^{-1}: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\cos\left(-\frac{\pi}{2}\right)$$

$$= \boxed{0}$$

$$14) \sin^{-1}\left(\sin\left(\frac{17\pi}{6}\right)\right)$$

$$\sin^{-1}: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$$

$$= \sin^{-1}(\sin(\frac{\pi}{6}))$$

$$= \boxed{\frac{\pi}{6}}$$

$$15) \tan^{-1}(\tan(\frac{5\pi}{6}))$$

$$\tan^{-1}: (-\frac{\pi}{2}, \frac{\pi}{2}) \xrightarrow{+2\pi}$$

$$\tan^{-1}(-\tan(\frac{\pi}{6}))$$

$$= \tan^{-1}(-\frac{1}{\sqrt{3}})$$

$$= \boxed{-\frac{\pi}{6}}$$

$$16) \sin(\sec^{-1}(-\sqrt{2}))$$

$$\sec^{-1}: [0, \pi] \text{ without } \frac{\pi}{2}$$

$$\sec \theta = -\sqrt{2}$$

$$\frac{1}{\cos \theta} = -\sqrt{2}$$

$$\cos \theta = -\frac{\sqrt{2}}{2}$$

$$\theta = \frac{3\pi}{4}$$

$$\sin(\frac{3\pi}{4})$$

$$\boxed{\frac{\sqrt{2}}{2}}$$

$$17) \text{ Rewrite the expression as an algebraic expression in } x$$

$$\sec(\tan^{-1}(\frac{x}{4}))$$

$$\begin{array}{c} \sqrt{x^2+16} \\ \diagup \quad \diagdown \\ 1 \quad x \end{array} \rightarrow \begin{array}{l} x^2 + 16 = r^2 \\ r = \sqrt{x^2+16} \end{array}$$

$$\boxed{\sqrt{x^2+16}}$$

$$18) \text{ Rewrite the expression as an algebraic expression in } x$$

$$\cos(\sin^{-1}(\frac{x}{4}))$$

$$\begin{array}{c} 4 \\ \diagdown \quad \diagup \\ x \quad \sqrt{16-x^2} \end{array} \rightarrow \begin{array}{l} x^2 + z^2 = 16 \\ z = \sqrt{16-x^2} \end{array}$$

$$\boxed{\sqrt{16-x^2}}$$

$$19) \text{ Find functions } f \text{ and } g \text{ such that } F = f \circ g$$

$$F(x) = (\tan^{-1}(x))^2$$

$$\{f(x), g(x)\} = \boxed{\{x^2, \tan^{-1}(x)\}}$$